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Q1. Operators in Python

ANSWER

operators are symbols which is used to calculate two operands it either returns a value or returns true or false

1. Arithmetic operators
 2. Comparison operators
 3. Logical operators
 4. Assignment operators
 5. Membership operators
1. Arithmetic operators is used to calculate the operands
Addition(+), subtraction(-), multiplication(*), division(/)(it returns the quotient), modulus%(it returns the remainder)
it returns
2. comparison operators is used to compare the two operands
greater than(>), less than(<), greater than or equal to(>=), less than or equal to(<=) etc
3. logical operator is used to combine conditions. it also returns the value true or false
and, or, not are the operators
4. assignment operator is the "=" operator which assigns the value in the right to the variable in the left
a=4
here the value 4 is assigned to the variable a
5. membership operators are "in" and "not in"
this is used to check whether the given operand is in the specific string or not

Q2. Operators

ANSWER

1. division operator
it divides and returns a value. it returns the quotient value. It returns floating values
eg $5 / 2 = 2.5$
2. floor division operator
Divides and returns the quotient but it either returns the integer part or it rounds off the value and returns it
Eg $5 // 2 = 2$
3. modulus
it returns the remainder after division
Eg $5 \% 2 = 1$
4. exponential
it considers the first operand as the base and the second operand as the power and calculates it
Eg $2 ** 3 = 8$
here the base is 2 and the power is raised to 3 so 2 cube is 8

Q3. for and while loop

ANSWER

1. The for loop iterates the statement inside the loop repeatedly until the condition or the range exceeds
2. the for loop is used for fetching characters from string, list, tuple etc
3. the key difference between for and while loop is that in for loop the number of iterations is known but in while loop the number of iterations is based on the condition given in the while loop. it runs as long as the condition is false

Q4. range()

ANSWER

1. The range function is used in for loop to determine the number of iterations
 2. it consists of range(start, stop, step)
- i. range(5)=it starts from 0 and the stop value is always the given number minus one so the output will be
0, 1, 2, 3, 4
- ii. range(1, 6)=it starts from 1 and stops in 6-1=5
1, 2, 3, 4, 5
- iii. range(1, 10, 2)=it starts from 1 and stops at 10-1=9 and here the step value is included which is 2 that is it prints every second iteration
1, 3, 5, 7

Q5. String

ANSWER

1. string is a sequence of characters enclosed within single quotes ' ' double quotes " " or triple quotes " " "
2. strings are immutable that is once a string is created it cannot be modified
3. if a string has to be modified python creates a new string and the old string remains the same

Q6. String Slice

ANSWER

```
ytho  
nohtyp  
pto  
htyp
```

Q7. List

ANSWER

- 1.list is the collection of values enclosed within square brackets.
- 2.list is mutable string is immutable
- 3.in list element can be different data type where as in strings all these are characters

Q8. List Slice

ANSWER

```
[20,30,40,50,60]  
[10,20,30,40]  
[10,30,50,70]  
[70,60,50,40,30,20,10]  
[40,50,60]
```

Q9. Index

ANSWER

- 1.The characters in the strings are counted and fetched by the index numbers
2. the index number starts from 0 and ends in n
- 3.the reverse or negative indexing starts from -1 and decreases in the opposite order

Q10. length

ANSWER

- 1.len() function is used to find the length of the string and it also counts the space
- 2.count() function is used to count the number